CBS COPENHAGEN BUSINESS SCHOOL

TEXT CLASSIFICATION

Naive Bayes Classification

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Agenda

Introduction

What is Text Classification?

Bayes' Theorem

Naive Bayes Classifier

Implementing Naive Bayes

Live coding walkthrough

Model Evaluation

Precision

Recall

F-measure

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Can you think of examples where classification could be useful in your daily lives?

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Text Classification

Process of automatically categorizing text documents into predefined categories or classes based on their content.

Classification Methods

Supervised Techniques

Trains on labelled data to predict outcomes

- Naive Bayes
- Logistic Regression

Unsupervised Techniques

Discovers patterns in unlabeled data

- Latent Dirichlet Allocation (LDA)
- K-Means Clustering

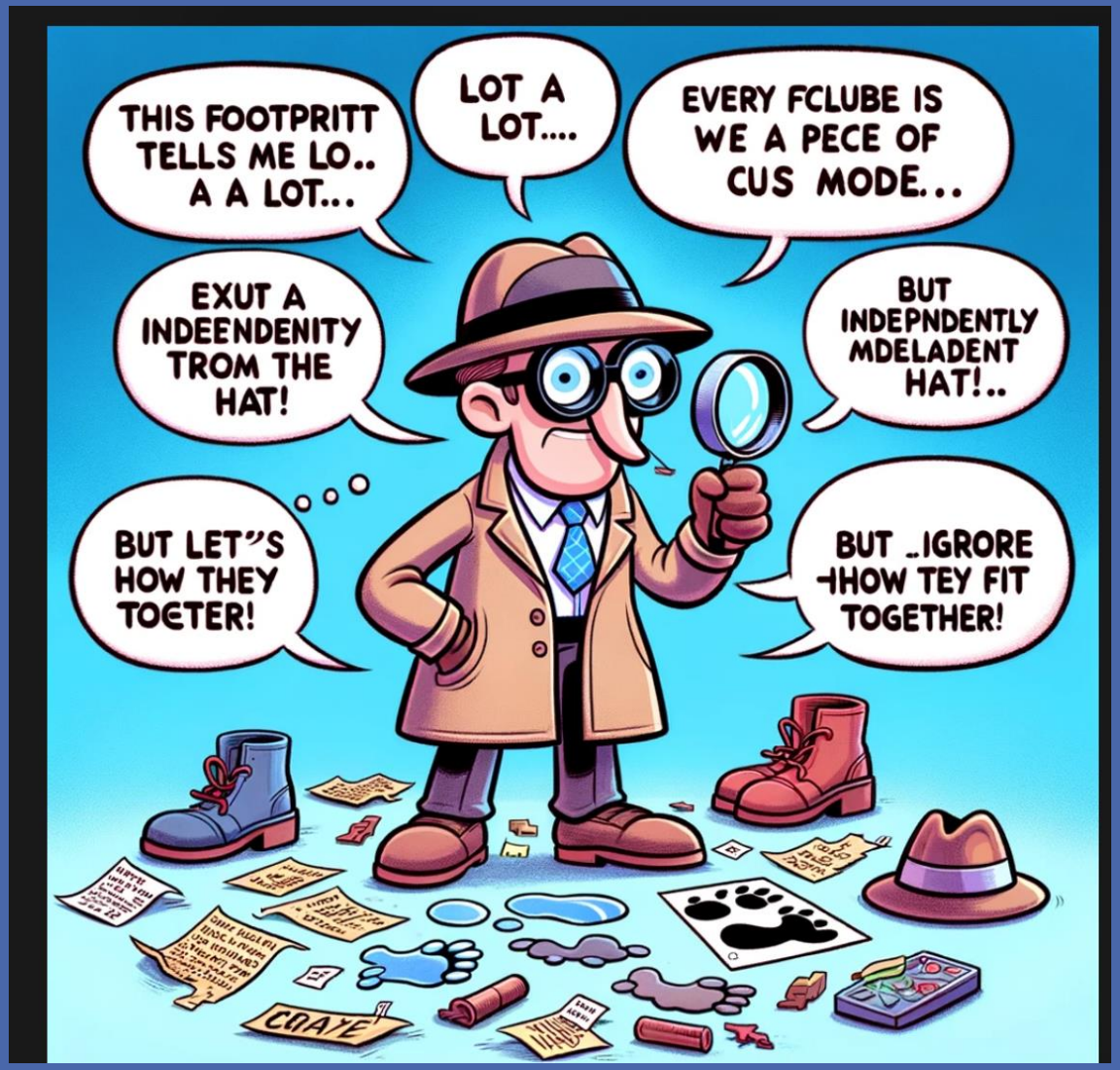
Deep Learning Techniques

Uses neural networks for complex pattern recognition in both labeled and unlabeled data

- RNNs, LSTM
- BERT, GPT

Detective Naive Bayes

- A simple probabilistic classifier based on Bayes' theorem.
- Use to make predictions based on the probabilities of observing certain features in the data.
- Strong (naive) independence assumptions between the features.



Solving mysteries - one independent clue at a time!

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Bayes' theorem is used to:

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What does the term 'prior probability' refer to?

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What does the 'posterior probability' indicate?

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Bayes' Theorem

How to update the probabilities of hypotheses when given evidence?

Prior

What we believed before

P(H) - Prior Probability of hypothesis H before observing the evidence E.

Likelihood

How well the hypothesis explains the data

P(E|H) - Likelihood of observing evidence E given hypothesis H.

Evidence

How common the data is

P(E) - Probability of observing the Evidence E.

Posterior

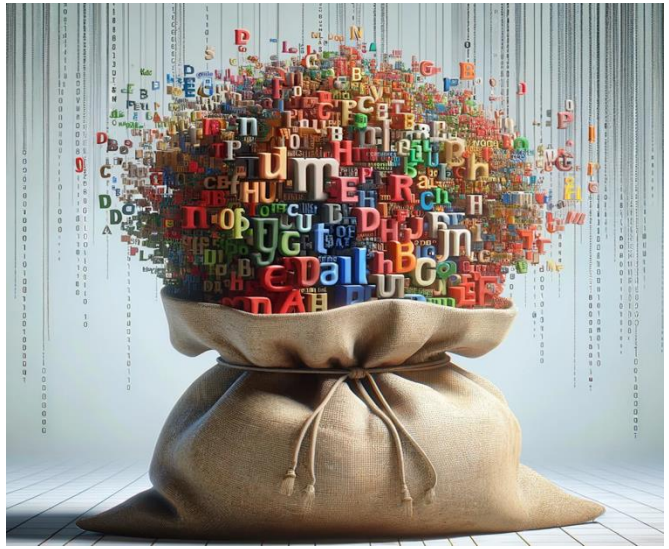
What we believe after

P(H|E) - Posterior probability of hypothesis H given evidence E.

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

Feature Extraction: Bag of Words

Imagine your favorite book.
Now, tear out all the pages, cut out each word, and toss them into a big bag.
What do you have ?



Treat text as a collection of unconnected words

Ignore grammar and word order

Count word frequencies

Word frequencies are used as a feature

Naive Assumption

Each feature contributes independently to the probability of the class label, regardless of the presence or absence of other features.

If we classify the fruits based on three features: Colour, Shape and Taste

Red, spherical, sweet

Orange, oval, sweet-tart

Naive Bayes for text classification

Find the most probable class c for a given text document d

Document d represented as words $w_1 \ldots w_n$

Most likely class -> Maximum a Posteriori (MAP)

Baye's Theorem: $P(c|d) = \frac{P(d|c)P(c)}{P(d)}$

Since P(d) is constant for all classes drop the denominator

$C_{MAP} = \text{argmax}_{c \in Class} P(d|c).P(c)$ = likelihood . Prior

Given the Naive assumption of independence between words:

$C_{MAP} = \text{argmax}_{c \in C} P(w_1, w_2 \ldots w_n|c).P(c) = P(w_1|c).P(w_2|c) \ldots P(w_n|c).P(c)$

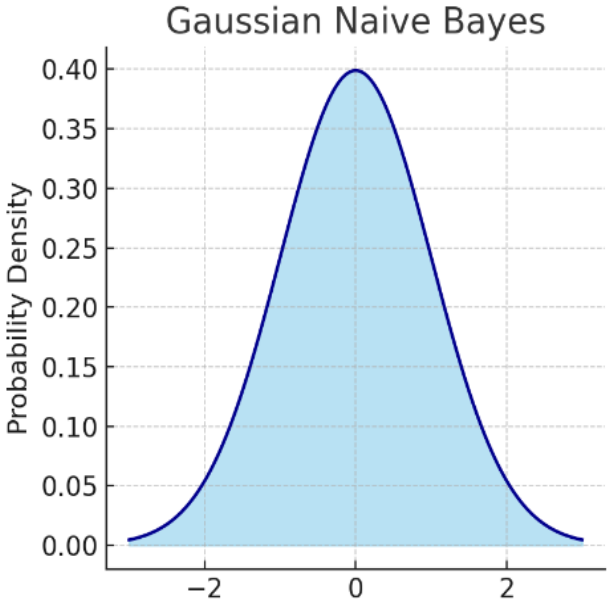
Types of Naive Bayes Models

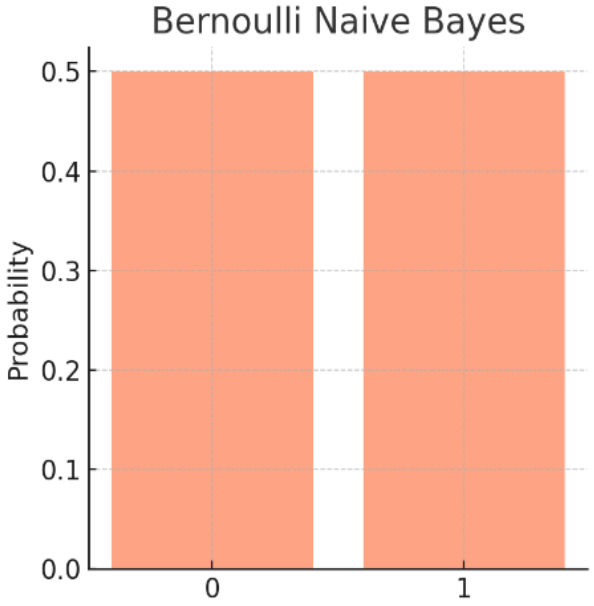
Gaussian: features follow a normal distribution

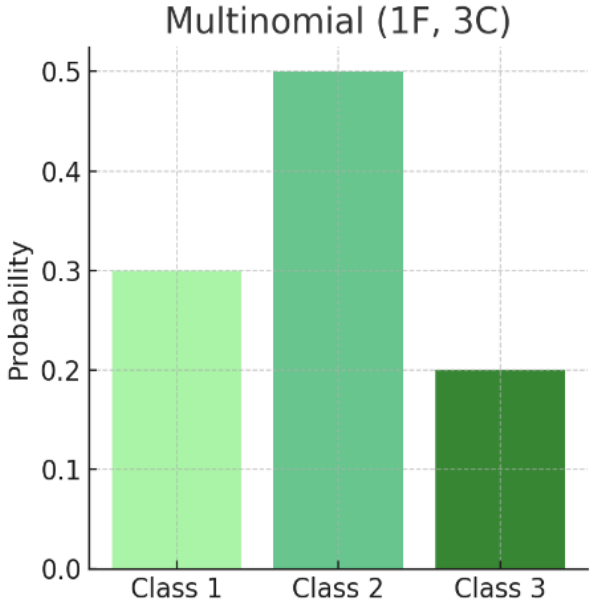
Bernoulli: Binary features (0/1, Yes/No)

Multinomial: Best suited for feature counts

- features are continuous and normally distributed.
- E.g. words present or not is the feature.
- E.g. Word counts or frequencies in documents.







Implementing Naive Bayes

Sentiment Classification

1. Get a corpus of tweets, clean and pre-process the data
2. Divide new data into positive and negative class (use afin score)
3. Compute frequency of word for both class P(word l pos), P(word|neg)
4. Get likelihood $\lambda(w) = P(\text{word}|\text{class})$ and prior = $P(\text{class})$
5. Test the classifier (predict by using "λ" dictionary)

Naïve Bayes as a Language Model

Which class assigns the higher probability to sentence?

Learn word probabilities for each class

- $P(\text{word}|\text{class})$

Score a sentence under a class

- $\text{P}(\text{sentence}|\text{class}) = \prod_i P(\text{word}_i|\text{class})$

Apply Bayes' rule

- $\text{P}(\text{class}|\text{sentence}) = P(\text{sentence}|\text{class}) * P(\text{class})$

Prediction:

- Choose the class with the highest probability.

Evaluation

Accuracy?

If model predict all cancer-free correctly

- 90 patients

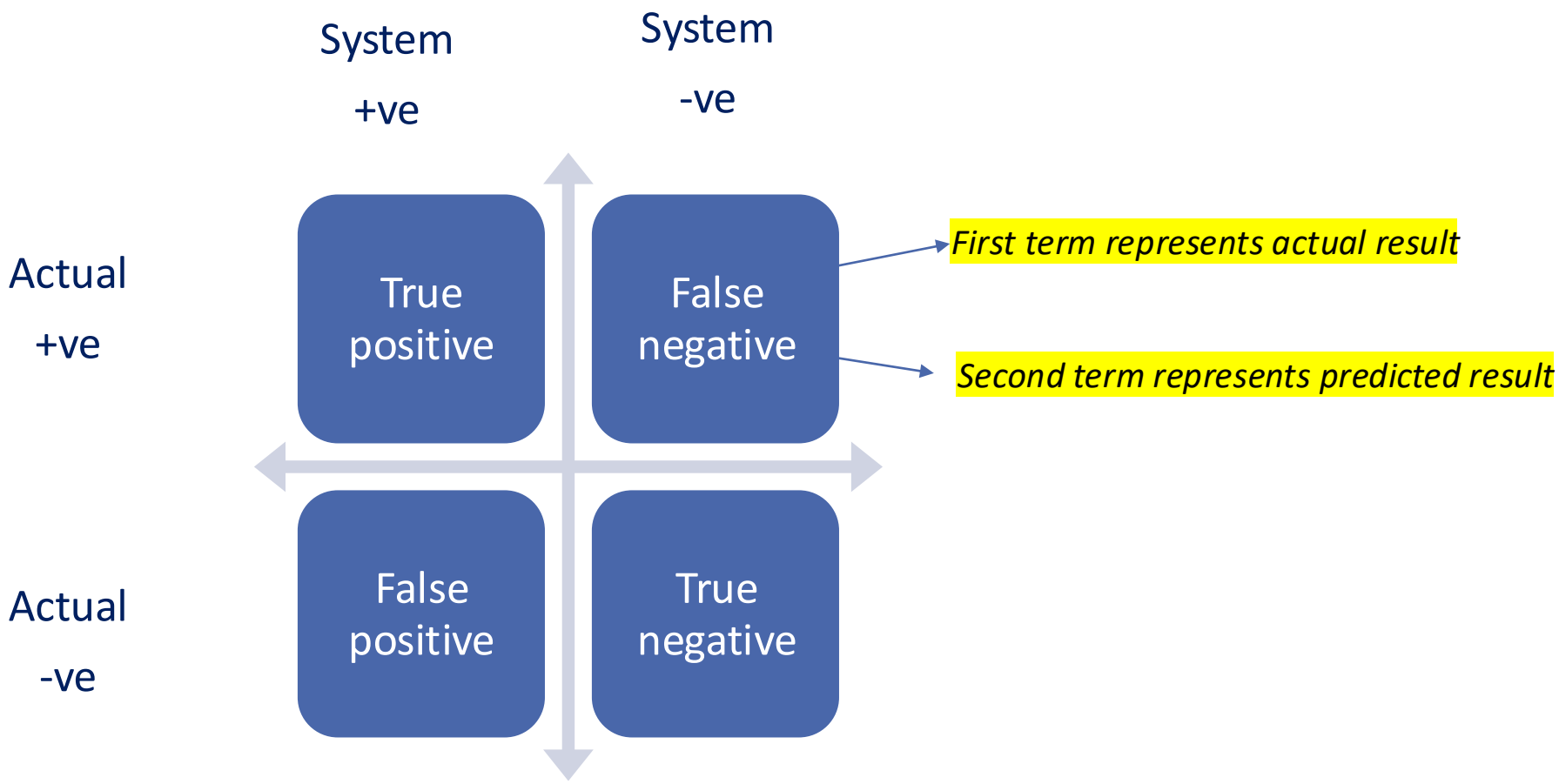
or completely misclassified all cancer patients

- 10 patients

Still high accuracy

- Accuracy= $\frac{90}{100}$ =90%

Confusion Matrix for Evaluation



Example: Confusion Matrix

90 patients are cancer-free

- 10 patients have cancer

Model identified **8** patients as positive (cancer)

- Only **6** had cancer
- **2** were cancer-free

Model labelled **92** patients as negative

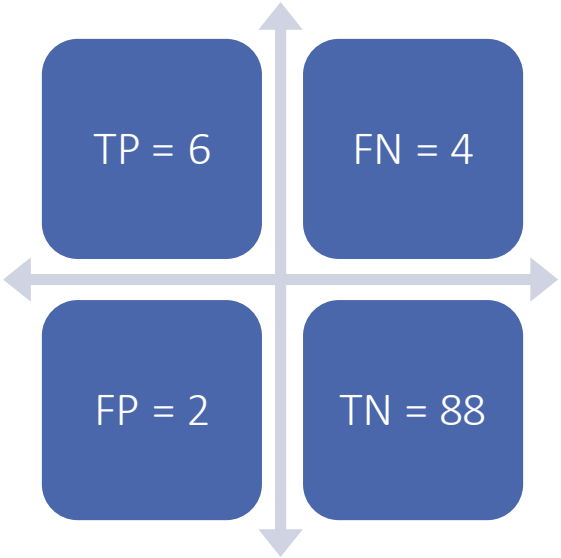
- **88** were cancer-free
- Missed **4** cancer cases

True Positives (TP): Cancer patients correctly identified as having cancer..

False Negatives (FN): Cancer patients incorrectly identified as cancer-free.

False Positive (FP): Healthy patients incorrectly identified as having cancer.

True Negatives (TN): Healthy patients correctly identified as cancer-free.

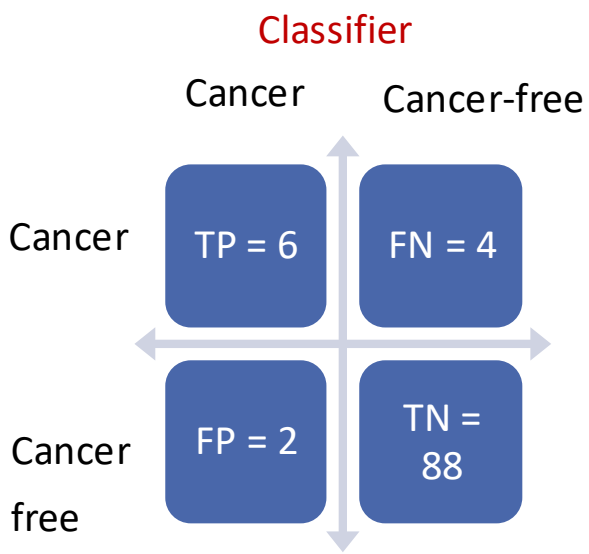


Precision (Positive Predictive Value)

When the test predicts cancer, how often is it correct?

Correctly identified cancer patients divided by all patients predicted as having cancer.

- Precision $= \frac{\text{True positive}}{\text{True positive + False positive}} = \frac{6}{6+2} = \frac{6}{8}$

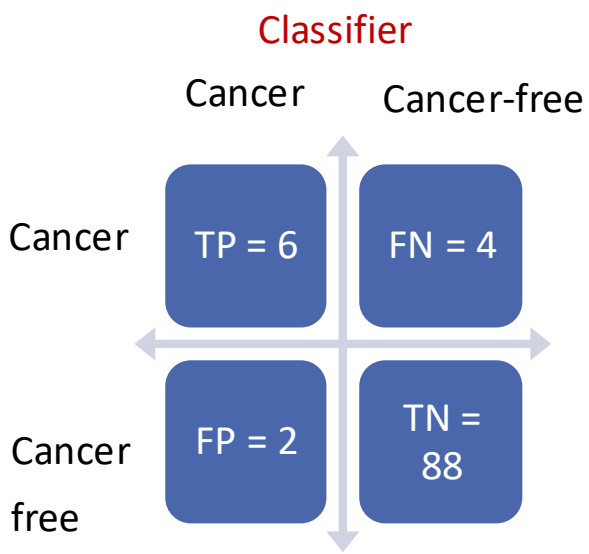


Recall (Sensitivity)

When a patient actually has cancer, how often does the test detect it?

Correctly identified cancer patients divided by all patients who actually have cancer.

- Recall $= \frac{\text{True positive}}{\text{True positive + False negative}} = \frac{6}{6+4} = 0.6$



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High precision and low recall means:

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F - MEASURE

- A number that combines Precision and Recall

$$F_\beta = \frac{(\beta^2+1)\text{Precision} \times \text{Recall}}{\beta^2\,\text{Precision}+\text{Recall}}$$

- Common practice to use balanced ($\beta = 1$)

- $F_1 = \frac{2\times\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}}$

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Why Naive Bayes?

- Simple
- Scalable
- Efficient especially when working with large data
- Requires a relatively small amount of training data
- Ignore the missing values in probability calculation

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Why Not Naive Bayes?

Independence Assumption

Rarely true

May lead to incorrect predictions when features are correlated.

Zero Frequency Problem

If a document has a class in **test data** but not in training data, it will assign a 0 probability.

Probability Estimation Bias

Can be biased towards majority class or more feature occurrences.

This also effect accuracy.

Feature Relevance

Problematic if some features are irrelevant or misleading.

Key Takeaways

- Naive Bayes is computationally efficient, making it suitable for large datasets and real-time analysis.
- Easy to implement and interpret, treat as baseline model.
- Uses probability estimates for predictions, which can be valuable for decision-making processes.
- Capable of handling both binary and multiclass classification problems.

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What is a common issue when using Naive Bayes on imbalanced datasets?

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